

M6125 — Quick Guide to the Three Hose Lay Types

FWFD uses three water-supply lay tactics: **Reverse Lay**, **Laying Dry**, and **Forward Lay**. All three aim for the same end state — a volume connection at the hydrant feeding the manifold engine — they just differ in *which engine ends up at the hydrant and how it gets there*.

1. Reverse Lay

When: Preferred method for all apartment and commercial fires.

Who: The supply engine (typically 2nd-in) lays *from the scene to the hydrant*.

How it works:

1. Supply engine stops near the manifold engine; supply line is placed under the manifold engine's tire (slack, not yet connected).
2. Supply engineer drives to the hydrant — **not exceeding 10 mph** — counting LDH sections laid.
3. Manifold engineer connects the LDH to intake once tension is off the line.
4. At the hydrant, the supply engineer:
 - Engages the pump before exiting the cab
 - Sends tank water first (notify manifold engine — tank water only)
 - Attaches threaded-to-Storz adaptor to the steamer + closed 2½" gate valve to the 2½" outlet
 - Attaches LDH from steamer to intake, opens hydrant, bleeds air, opens intake valve
 - Attaches a second LDH from the gate valve to the opposite intake → **volume connection**
 - Marks static pressure on the gauge, coordinates flow/pressure needs with manifold engine
 - Maintains minimum 20 psi residual intake pressure

Why it's preferred: Most efficient way to establish a volume connection; puts an engine (and 2 extra firefighters) at the hydrant for use on other assignments; doesn't require the Humat valve.

2. Laying Dry

When: Used when a reverse lay isn't feasible, but you still want the reverse-lay advantages (apartment/commercial fires).

Who: Two engines split the job — **2nd-on** lays the dry line, **3rd-on** wraps the hydrant.

How it works:

- **2nd-on Engine:**

1. Stops at the hydrant; firefighter wraps the supply line around the hydrant (slack) and re-boards
2. Lays *from the hydrant to the manifold engine* (≤ 10 mph), counting LDH sections
3. Announces on radio: **"Laying Dry"** + hydrant location
4. Positions at scene so the line can be disconnected and hand-carried to the manifold engine (leave room for aerial placement)
5. Connects to the manifold engine for a tank-to-tank transfer
6. Crew now free for other assignment (e.g., search, back-up line)

- **3rd-on Engine:**

1. Reports to the wrapped hydrant
2. Follows the same hydrant hookup steps as the Reverse Lay Supply Engine (steamer adaptor, gate valve, second LDH → volume connection)
3. Stays with the apparatus, coordinates supply with manifold engine

Why it matters: Same end result as a reverse lay (volume connection, apparatus at hydrant) — just achieved with two engines instead of one when a true reverse lay isn't possible.

3. Forward Lay

When: Used when a reverse lay isn't practical/possible — typically adequate for one- or two-family residential fires, or when the 2nd-in engine is substantially delayed (in which case the 1st-in engine lays its own line).

Who: Supply engine lays *from the hydrant to the scene*.

How it works:

2. Supply engine stops at the hydrant; firefighter grabs the hydrant bag, **Humat valve**, and gate valve
3. Wraps supply line around hydrant (slack) — do not connect until tension is off
4. Notifies engineer it's safe to drive to the incident (≤ 10 mph)
5. At the hydrant: connects Humat valve + supply line to the steamer, connects closed 2½" gate valve to the 2½" discharge
6. Engineer positions at scene so the line can be disconnected and carried to the manifold engine
7. Manifold engineer requests water from the firefighter waiting at the hydrant; bleeds air, opens intake
8. Marks static/residual pressure; maintains 20 psi minimum residual

Augmenting later: A later-arriving engine can reinforce the forward lay via a **volume connection** — identified on radio as "**Water Supply**" — using LDH pigtails through the Humat valve into an additional intake.

Key difference from the other two: Forward Lay is the only one that **requires the Humat valve**.

Quick Comparison

	Direction	Engine ends up at hydrant?	Humat valve?	Best for
Reverse Lay	Scene → Hydrant	Yes (1 engine)	No	Apartments/commercial
Laying Dry	Hydrant → Scene (dry line)	Yes (3rd-on)	No	Apartments/commercial, when reverse lay isn't possible
Forward Lay	Hydrant → Scene	No (unless augmented)	Yes	1-2 family residential

Universal rules across all three:

- Never exceed **10 mph** while laying line
- Never connect a line under tension — relax it first (snags can damage pump/hydrant)
- Maintain minimum **20 psi** residual intake pressure; report to Command if approaching that limit

- Max discharge pressure: **300 psi** (woven-jacketed hose) / **200 psi** (rubber-jacketed hose)